QP CODE : 25901005

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Discipline Specific Core Course (DSC)-MG1DSCZIM101

INTRODUCTION TO MICROBIAL WORLD

Duration: 1.5 Hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part **A**

Fill in the blanks

Answer All Questions

Each Question carries **1** mark

1.Robert koch is the father of …… Microbiology **[K]/(CO1]**

Answer :Modern Microbiology

2. Cysts enable bacteria to survive adverse …… **[U]/[CO2]**

Answer : Condition

3 Zygomycota includes …….. **[K]/(CO4)**

Answer : Molds

4. Oncogenic RNA viruses can cause cancer through …… **[U]/[CO3]**

Answer : Integration

5. Algae are used as a source of …… **[U]/[CO4]**

Answer : Food

6. Study of Microorganism is …….. **[U]/[CO1]**

Answer : Microbiology

7. Viral…… facilitate attachment to host cells **[K]/[CO2]**

Answer : Protein

8. The RNA is surrounded by a…….. envelope **[U]/[CO3]**

Answer :Protein envelop

9. Extremophiles are microorganisms that thrive in……..environments [K]/[CO1]

Answer : Extreme

10. Sexual reproduction in protozoa involves the exchange of genetic material through the process of..... [K]/[CO5]

Answer : Conjugation

[1x10=10]

Part –B

Short Questions

Answer any **10** question

Each question carries **2** marks

11. Write a short note on Prochlorophyceae. [U]/[CO4]

Prochlorophyceae is a class of prokaryotic photosynthetic organism (Cyanobacteria) that contain chlorophylls a and b but lack phycobiliproteins which is unusual for cyanobacteria .They are found in marine and freshwater environments and are significant primary producers.

12. Distinguish the feature of Cytoplasmic membrane. [U]/[CO2]

Answer: The cytoplasmic membrane is a selectively permeable barrier. It regulates the passage of substances into and out of the cell, maintains internal homeostasis, and is primarily composed of a phospholipid bilayer with embedded proteins.

13. Write short note about cultivation of plant virus .[A]/[CO3]

Answer: Plant viruses are obligate intracellular parasites and thus cannot be cultivated on artificial media. They are typically grown in living plant hosts, tissue cultures, or protoplasts, which provide the necessary cellular machinery for replication.

14. Describe the features of Tribophyceae and Dinophyceae. [K]/[CO2]

Answer: Tribophyceae (yellow-green algae) primarily use chlorophyll a and c and store energy as oil, lacking phycobilins. Dinophyceae (dinoflagellates) are characterized by two flagella and cellulose plates, often containing chlorophyll a and c and sometimes causing red tides.

15. Explain the function of Flagella. [K]/[CO2]

Answer: Flagella are whip-like appendages primarily used for cell motility or locomotion. They enable movement in both prokaryotic and eukaryotic cells, though their structure and mechanism of action differ between the two types.

16. Explain main replication cycle in Bacteriophage. [K]/[CO3]

Bacteriophages typically undergo either a lytic cycle, where they replicate and lyse the host cell, or a lysogenic cycle, where their genetic material integrates into the host genome. The lytic cycle involves attachment, penetration, synthesis, assembly, and release of new virions.

17. Compare the similarities of Prokaryotic and Eukaryotic cell. [U]/[CO1]

Both prokaryotic and eukaryotic cells share fundamental similarities, including the presence of a cell membrane, cytoplasm, ribosomes, and DNA as their genetic material. Both types also perform basic life processes such as metabolism and reproduction.

18. Define golden age of Microbiology. [U]/[CO1]

The "golden age of Microbiology" refers to the period from the mid-1800s to the early 1900s, during which numerous fundamental discoveries were made. Key figures like Louis Pasteur and Robert Koch established microbiology as a science, identifying many pathogens and developing techniques like pasteurization and germ theory.

19. Draw and label the part of Archaebacteria. write down the function. [K]/[CO1]

Archaebacteria (Archaea) possess unique cell envelopes and structures, including a cell wall (lacking peptidoglycan) and a cell membrane. The cell membrane regulates transport, and functions like metabolism, growth, and reproduction occur within the cytoplasm.

20. What is HIV and how does it affect humans? [K]/[CO3]

HIV (Human Immunodeficiency Virus) is a virus that targets and destroys immune system cells, specifically CD4+bold cap C bold cap D 4 plus $\mathbf{CD4}$+T cells. This leads to a compromised immune system, eventually causing Acquired Immunodeficiency Syndrome (AIDS), making the body vulnerable to opportunistic infections.

21. Write a short note on Viroids. [K]/[CO3]

Viroids are the smallest known infectious agents, consisting of a single strand of circular RNA without a protein coat. They primarily infect plants, causing various diseases by interfering with normal gene expression and replication processes.

22. Write short on the basis of economic important of Zygomycota and Deuteromycota [U]/[CO4].

Zygomycota are economically important for producing industrial enzymes and some food products like tempeh, though they also cause food spoilage. Deuteromycota (imperfect fungi) are crucial for producing antibiotics like penicillin and useful enzymes, but also cause significant plant and animal diseases.

[2x10=20]

Part – C

Short Essay Question

Answer any **5** Questions

Each Question carries **4** Marks

23. Discuss the ecological significance of Cyanobacteria with suitable diagram. [A]/[CO4].(Significance -2 Marks, Diagram -2 marks)

Cyanobacteria are crucial primary producers and nitrogen fixers. They perform oxygenic photosynthesis, producing oxygen and forming the basis of many food chains. They also fix atmospheric nitrogen into usable forms like ammonia, enriching soil and water.

24. Illustrate the ultrastructure of bacteria and their functions. [A]/[CO2].(Diagram -1 Mark, Explanation -2 Marks, Function -1 Mark)

Bacteria have a simple prokaryotic ultrastructure including a cell wall, membrane, cytoplasm, ribosomes, and a nucleoid. Key features are the absence of membrane-bound organelles and a true nucleus. Functions include protection (cell wall), selective transport (membrane), protein synthesis (ribosomes), and genetic information storage (nucleoid).

25. Discuss the role of Saccharomyces in industrial field. [A]/[CO4].(Any 4 Industrial Application - 1mark each)

Saccharomyces is primarily used in the fermentation industry. Saccharomyces cerevisiae is a type of yeast crucial for producing alcoholic beverages like beer and wine through fermentation of sugars. It is also widely used in baking as a leavening agent to make bread rise by producing carbon dioxide gas. The yeast converts carbohydrates into ethanol and carbon dioxide.

26. Differentiate the Economic important of Ascomycota and Basidiomycota. [K]/[CO4]. (Economic Importance of Ascomycota -2 Marks, Economic important of Basidiomycota -2 Marks)

Ascomycota and Basidiomycota have distinct economic roles, primarily in food production and decomposition .Ascomycota includes beneficial organisms like Saccharomyces (brewing/baking) and edible truffles, but also plant pathogens like Dutch elm disease. Basidiomycota are mainly known for edible mushrooms (like button mushrooms) and are significant decomposers of wood, playing a vital role in nutrient cycling, though some are also plant pathogens.

27. Discuss the morphological feature of Bacterial cell wall. [A]/[CO2] (Differentiate Gram positive and Gram Negative bacterial cell wall -2 Mark each)

Bacterial cell walls are primarily composed of peptidoglycan and provide shape and structural integrity.The cell wall is a rigid layer outside the plasma membrane. Gram-positive bacteria have a thick peptidoglycan layer, while Gram-negative bacteria have a thin peptidoglycan layer and an outer membrane containing lipopolysaccharides. This structure protects the cell from osmotic lysis and determines the bacterial cell's shape.

28. Explain the detailed note on Bacteriophage. [K]/[CO3]. (Define Bacteriophage -1 mark , Life Cycle – 2 Marks, Diagram -2 Marks)

A bacteriophage is a virus that specifically infects bacteria .Bacteriophages, or phages, are composed of a protein coat enclosing genetic material (DNA or RNA). They replicate by inserting their genome into the host bacterium, either through a lytic cycle (killing the host cell) or a lysogenic cycle (integrating into the host genome). They are widely studied in molecular biology and have potential applications in phage therapy.

29. Compare the characteristic feature of Aspergillus and Penicillium. [U]/[CO4].(Characteristics of Aspergillus - 2 Marks , Characteristics of Penicillium – 2 Marks)

Aspergillus and Penicillium are both filamentous fungi with distinct conidial structures and economic importance. Aspergillus species are known for producing toxins (like aflatoxin) and causing diseases, but some are used in fermentation. Penicillium is famous as the source of the antibiotic penicillin and is used in cheese making. Morphologically, Aspergillus typically has a single row of phialides on a vesicle, while Penicillium has a brush-like (penicillus) arrangement of conidiophore.

[4x5=20]